

Md Kamrujjaman Mobin

Software Engineer I, Samsung R&D Institute, Dhaka, Bangladesh.

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RESEARCH INTERESTS

Computer Vision. Graphical 3D Modeling, Vision Language Models.

EDUCATION

Shahjalal University of Science and Technology, Sylhet, Bangladesh

Feb 2020 — July 2025

BSc. in **Computer Science and Engineering**

Cumulative GPA: **3.83**/4.00

Thesis Title: AI-Driven Health System for Mortality Prediction Using Electrocardiograms (ECGs) with Transformers and Selective State-Space Models.

PROFESSIONAL EXPERIENCE

Samsung R&D Institute Bangladesh

Dhaka, Bangladesh

Software Engineer I

July 2025 — Present

- Developed core utility applications for **Samsung Galaxy Smartwatch 7 & 8** (Java, Wear OS), integrating and calibrating sensor subsystems including gravity, magnetic, gyroscope, accelerometer, and vector sensors.
- Improved maintainability and code quality by elevating **SAM** (Structure, Abstraction, Modularity) score by **7.5%**, reducing **CBO** (Coupling Between Objects) by **20%**, achieving **>90%** unit test coverage, and leading code reviews with enforced clean code standards to ensure consistency and readability across the codebase.
- Spearheading R&D initiatives in **AI-driven audio/video processing** and **non-invasive blood glucose estimation** using wearable Photoplethysmography (PPG) sensing.
- Contributing to a global smartwatch ecosystem serving **450M+** active smartwatch users worldwide (2025).

PUBLICATIONS

Kamrujjaman Mobin, Md & Saiful Islam, Md & Al Barid, Sadik & Masum, Md. "Cardioformer: Advancing AI in ECG Analysis with Multi-Granularity Patching and ResNet", The 38th Canadian Conference on Artificial Intelligence (Canadian AI 2025), Calgary, Canada, 2025, DOI: [10.48550/arXiv.2505.05538](https://doi.org/10.48550/arXiv.2505.05538). Source Code: github.com/KMobin555/Cardioformer. (**Accepted, In Progress**)

Kamrujjaman Mobin, Md. "Transforming ECG Images to Waveforms with UNet-LSTM and Multilabel Classification with ResNet-LSTM: A Multimodal Approach by PulsePlex", Computing in Cardiology 2024 (CinC), Karlsruhe, Germany, 2024. DOI: [cinc.org/archives/2024/pdf/CinC2024-393](https://doi.org/10.1109/CinC2024-393). Source Code: github.com/KMobin555/George-B-Moody-Physionet-2024. (**Published**)

M. K. Mobin, S. Al Barid and M. S. Rahman, "CricPose360: A Comprehensive Dataset for Real-Time Pose-Enhanced Cricket Shot Classification Using YOLOv8," The 27th International Conference on Computer and Information Technology (ICCIT), Cox's Bazar, Bangladesh, 2024, pp. 563-568, DOI: [10.1109/ICCIT64611.2024.11022536](https://doi.org/10.1109/ICCIT64611.2024.11022536). Source Code: github.com/KMobin555/CricPose360. (**Published**)

Kamrujjaman Mobin, Md & Saiful Islam, M. "LuxVeri at GenAI Detection Task 1: Inverse Perplexity Weighted Ensemble for Robust Detection of AI-Generated Text across English and Multilingual Contexts", The 31st International Conference on Computational Linguistics (COLING), Dubai, UAE, 2025. DOI: [10.48550/arXiv.2501.11914](https://doi.org/10.48550/arXiv.2501.11914). Source Code: github.com/KMobin555/Luxveri-at-GenAi-Detection-Coling-2025. (**Published**)

Kamrujjaman Mobin, Md & Saiful Islam, M. "LuxVeri at GenAI Detection Task 3: Cross-Domain Detection of AI-Generated Text Using Inverse Perplexity-Weighted Ensemble of Fine-Tuned Transformer Models", The 31st International Conference on Computational Linguistics (COLING), Dubai, UAE, 2025. DOI: [10.48550/arXiv.2501.11918](https://doi.org/10.48550/arXiv.2501.11918). Source Code: github.com/KMobin555/Cross-Domain-MGT-Detection-at-Coling-2025. (**Published**)

Ongoing Research

- Vision-Language foundation modeling to enhance downstream tasks in medical and visual reasoning.
- Eye-gaze analysis for detecting reading and scanning behavior using smart glasses for accessibility.

CONFERENCES & WORKSHOPS ATTENDED

- **52nd Computing in Cardiology Conference (CinC), 2025**, São Paulo, Brazil. Sep 2025
- **38th Canadian Artificial Intelligence Conference (Canadian AI), 2025**, Calgary, Canada. May 2025
- **31st International Conference on Computational Linguistics (COLING)**, Dubai, UAE. Jan 2025
- **51st Computing in Cardiology Conference (CinC), 2024**, Karlsruhe, Germany. Sep 2024

CO-CURRICULAR ACTIVITY

Machine Learning & Data Science Competitions Dhaka, Bangladesh
Kaggle Expert 2022 — Present

- **Image Matching Challenge 2024: Bronze Medal** (Top 6.3%). Designed a multi-model pipeline for large-scale 3D reconstruction. Integrated COLMAP with deep matchers (LightGlue, SuperGlue, DKM, LoFTR) for accurate correspondence estimation. [\[Notebook\]](#)
- **ISIC 2024 – Skin Cancer Detection: Bronze Medal** (Top 5.7%). Developed CNN + tabular feature models for melanoma classification. Used advanced augmentations (noise-based, mixup) to improve generalization and reduce overfitting. [\[Notebook\]](#)

Competitive Programming Online Platforms
Self-Directed Learning & Problem Solving 2020 — Present

- Solved **1500+ algorithmic problems** across major platforms. Active on Codeforces [\[Mr_Bin\]](#), CodeChef [\[mobin_555\]](#), and LeetCode [\[Mobin555\]](#). Demonstrated strong skills in algorithms, data structures, and optimization under time constraints.

SKILLS

- **Programming Languages:** Python, C++, Java, JavaScript, C.
- **Frameworks & Libraries:** PyTorch, Hugging Face, Scikit-learn, OpenCV, NumPy, Pandas.
- **Specialized Areas:** Large Language Models (LLMs), Natural Language Processing (NLP), Computer Vision, Biosignal Processing.
- **Soft Skills:** Research Writing, Academic Presentation, Problem-Solving, Collaboration, Critical Thinking.

REFERENCES

Md Saiful Islam

Data Scientist, Department of Computer Science, Athabasca University, Alberta, Canada

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Prof. Mohammad Shahidur Rahman, PhD

Professor, Computer Science and Engineering, Shahjalal University of Science and Technology, Sylhet, Bangladesh

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Scholar Profiles: [Personal Page](#) — [Google Scholar](#) — [linkedin.com/in/m-shahidur-rahman-b5b43020](https://www.linkedin.com/in/m-shahidur-rahman-b5b43020)

Prof. Md Masum

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Scholar Profiles: [Personal Page](#) — [IEEE Explorer](#)